

1 Counting Techniques

The following enumeration methods are drawn from *multiplication principle*.

1.1 Summary

Way to Sample	Sample You Get		Number of Samples
ordered sampling without replacement	a permutation of n objects		$n!$
	an ordered sample of size r	a permutation of n objects taken r at a time	${}_nP_r = \frac{n!}{(n-r)!}$
ordered sampling with replacement			n^r
unordered sampling without replacement	a distinguishable permutation	a combination of n objects taken r at a time	${}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$
		a distinguishable permutation of objects of m types	$\binom{n}{n_1, n_2, \dots, n_m} = \frac{n!}{n_1! \dots n_m!}$
unordered sampling with replacement			$\binom{n+r-1}{r}$

1.2 Some Notations

The expression ${}_nC_r$ can be read as “ n choose r ”.

The numbers $\binom{n}{r}$ are called **binomial coefficients**. They arise in the expansion of a binomial.

$\binom{n}{n_1, n_2, \dots, n_s}$ can be called a **multinomial coefficient**.

$$(x_1 + x_2 + \dots + x_s)^n = \sum_{\substack{n_1+n_2+\dots+n_s=n \\ n_1, n_2, \dots, n_s \in \mathbb{Z}_{\geq 0}}} \binom{n}{n_1, n_2, \dots, n_s} x_1^{n_1} x_2^{n_2} \dots x_s^{n_s}$$

1.3 Unordered Sampling with Replacement

The number of unordered samples of size r selected from n different objects with replacement is equal to the number of non-negative integer solutions to the equation

$$x_1 + x_2 + \dots + x_n = r \quad \text{with } x_i \in \{0, 1, 2, \dots, r\}, \quad i \in \{1, 2, \dots, n\}$$

where x_i represents the number of times object i is selected.

For each solution $(\hat{x}_1, \hat{x}_2, \dots, \hat{x}_n)$, we replace x_i with $\hat{x}_i$ vertical lines in the equation:

$$||| + +|| + \dots + || + = r$$

Then the number of solutions equals the number of permutations of r vertical lines and $n-1$ plus signs.

Therefore, the number of the original unordered samples is

$$\binom{n+r-1}{r} = \binom{n+r-1}{n-1} = \frac{(n+r-1)!}{r!(n-1)!}$$